

Advanced (Habanero)

- Q1.** What is the purpose of a two-way frequency table?
- (A) To display one categorical variable
 - (B) To display two numerical variables
 - (C) To analyze the relationship between two categorical variables
 - (D) To calculate means and medians
 - (E) To create a bar graph
- Q2.** A survey of 12 students finds that 8 like pizza and 7 like ice cream. If 3 students like both, what is the number of students who like only pizza?
- (A) 2
 - (B) 3
 - (C) 5
 - (D) 6
 - (E) 8
- Q3.** In a two-way table, the relative frequency of a cell is calculated as the cell frequency divided by the
- (A) row total
 - (B) column total
 - (C) table total
 - (D) grand total
 - (E) sample size
- Q4.** If $P(A) = 0.4$ and $P(B) = 0.6$, what is the probability of both events A and B occurring if they are independent?
- (A) 0.12
 - (B) 0.24
 - (C) 0.36
 - (D) 0.48
 - (E) 0.6
- Q5.** What type of association exists between two variables if the relative frequency of one variable increases as the other variable increases?
- (A) Negative association
 - (B) Positive association
 - (C) No association
 - (D) Inverse association
 - (E) Direct association
- Q6.** A study finds that there is no significant association between the amount of x and y . What can be concluded?
- (A) There is a negative association between x and y
 - (B) There is a positive association between x and y
 - (C) There is no causal relationship between x and y
 - (D) The sample size is too small
 - (E) The data is normally distributed
- Q7.** The marginal frequency of a row in a two-way table is the
- (A) sum of the frequencies in that row
 - (B) average of the frequencies in that row
 - (C) difference between the frequencies in that row
 - (D) product of the frequencies in that row
 - (E) ratio of the frequencies in that row
- Q8.** In a two-way table, if the relative frequency of one cell is 0.2, what percentage of the total frequency does that cell represent?
- (A) 10
 - (A) 15
 - (A) 20
 - (A) 25
 - (A) 30

Open Ended Questions (FRQ)

- Q9.** Create a two-way frequency table to display the relationship between favorite sport (football, basketball, or baseball) and favorite music genre (rock, pop, or hip-hop) for a sample of 15 students. Calculate the relative frequencies and interpret the results.
- Q10.** Suppose a company conducts a survey to determine the relationship between the level of education (high school, college, or graduate degree) and annual income ($< 30,000$, $30,000 - 50,000$, or $> 50,000$) for a sample of 20 employees. Create a two-way table and calculate the relative frequencies to analyze the association between these two variables.

Chef's Tips

- Tip 1: Ensure students understand the concept of relative frequencies
- Tip 2: Emphasize the importance of interpreting results in context
- Tip 3: Encourage students to create visual representations of the data